Table 1: Related studies: reported targets, data units and evaluation approaches.

Study	Response	Data unit	Evaluation
Wisniewski et al. (2020)	Next-hour signed-position interval	One market maker's hourly FX positions	Pinball and interval scores; coverage tests
Boudoukh et al. (2004)	Worst cumulative portfolio-return quantile	Fixed portfolio under a lognormal diffusion	MaxVaR/VaR ratios; discrete-monitoring simulation
van den Hoogen (2017)	Conditional quantile of cumulative payment-deficit peak	ABN AMRO transaction accounts	Coverage and independence backtests
This study	Sampled absolute-position peak	FX positions; weekly position aggregates	Pinball loss, breaches, model confidence sets and costs

Table 2: Data and matched six-record peak summaries.

Panel A. Sources, aligned samples and existing split dates

Data	Frequency	Series / records	Unit	Sample	Validation	Evaluation
FX, hourly	Hourly	1 / 18,576	USD mn	2014-02 to 2017-01	2015-10 to 2015-12	2016-01 to 2017-01
FX, daily	Day end	1 / 930	USD mn	2014-02 to 2017-01	2015-02 to 2015-04	2015-05 to 2016-12
Treasury bills	Weekly	1 / 706	USD mn	2013-04 to 2026-10	2019-07 to 2020-07	2021-01 to 2026-04
NY Fed panel	Weekly	12 / 706	USD mn	2013-04 to 2026-10	2019-07 to 2020-07	2021-01 to 2026-04
CFTC panel	Weekly	17 / 877	Contracts	2010-01 to 2026-10	2020-08 to 2021-08	2021-09 to 2026-09

Panel B. Within-series mean, sample SD and empirical 95th percentile

		$\lvert N \rvert$			Peak Y			Headroom Z		
Data	Unit	Mean	SD	95th	Mean	SD	95th	Mean	SD	95th
FX, hourly	USD mn	8.92	10.2	27.6	11.1	11.7	30.2	2.15	5.92	7.96
FX, daily	USD mn	9.00	10.5	27.5	17.9	16.6	41.0	8.91	14.8	24.9
Treasury bills	USD mn	39,500	27,200	92,600	55,300	30,400	108,000	15,800	16,000	50,000
NY Fed panel	USD mn	12,400	8,340	33,300	16,000	9,040	37,400	3,750	3,560	10,500
CFTC panel	Contracts	60,200	46,000	148,000	81,800	45,600	169,000	19,900	20,900	60,600

Notes. Records are per series; complete $H = 6$ peak counts are 18,570, 924, 700, 700 and 871, respectively, including the issue anchor. Earlier hourly FX validation spans 2015-04 to 2015-07 and evaluation spans 2015-07 to 2015-10; daily FX uses each nonempty source-naive day's last record, and both FX periods had been inspected. SD uses $m - 1$ and the empirical 95th percentile has rank $\lceil 95m/100 \rceil$; dollar values are in millions, panel entries are separate within-series medians without exposure pooling, and CFTC retains heterogeneous native contract counts.

Table 3: Forecast definitions and executed menus. Codes occur here to connect with the result files. Matched histories and update schedules make the reference explicit.

Method	Forecast definition	Executed menu / setting
Historical headroom (F)	Current floor plus a quantile of completed peak headrooms	Original FX histories {720,2880}; weekly {52,260}
Translated path (S)	Translate completed signed paths to the current position before taking their peaks	Same completed-label pools; fixed-window and rolling-window controls
Historical peak (K)	Historical absolute-peak quantile projected to the current floor	Same histories and update controls
Quantile regression (QR)	Training-scaled headroom regression with ridge slopes and an unpenalized intercept	Original three-point menu; initial panel five-point expansion; six-point menu $\{0.001, 0.01, 0.1, 1, 10, \infty\}$
Local quantiles (KNN)	Neighbor headroom quantile, with a separate increment-scale variant	Training-only transforms and fixed neighbor menus; Supplement Sections S8 and S23
Four-feature regression (QR4)	Current signed/magnitude positions and short/long increment RMS	Original five-point menu; six-point paired comparison below
Endpoint plug-in (L)	Floor plus a high absolute-displacement endpoint quantile	Proposition 2; empirical plug-in evaluated separately
Further benchmarks	Autoregressive replay; filtered paths; peak-tail fit; quantile forest	Existing bounded specifications; results and complete failures linked from Supplement Sections S7 and S34
Coverage and combinations	Delayed projected ACI-style wrappers; headroom SAV; fixed forecast combinations	Original-source wrapper/SAV inventory and broader combination inventory are separate

Training histories count completed labels. Fixed-window models refit every 24 FX records or 13 weekly records. Quantile and normalization conventions are detailed in Supplement Sections S8 and S23; this table retains the executed menus.

Table 4: Long-history short-horizon FX regression under two references and periods. Negative percentages favor regression. Matched gains recur while the rolling-reference direction changes.

Period / procedure	Reference	Loss change (%)	Issues	Strict breaches (regression / reference)
Later / original main comparison	Shorter-history rolling headroom	-12.66	6625	376 / 332
Later / matched	Same-history fixed-window headroom	-14.77	6625	376 / 360
Earlier / original structure	Shorter-history rolling headroom	+3.17	1440	See complete earlier-period tables
Earlier / matched	Same-history fixed-window headroom	-16.62	1440	20 / 16
Later / coverage-wrapped	Same-history wrapped headroom	-15.00	6625	329 / 328

Regression uses history 2880 and horizon 6; the original rolling reference uses 720. The earlier matched comparison is the original three-penalty regression, not the later one-standard-error selection rule. The wrapper comparison is a separate finite projected delayed ACI-style exercise.

Table 5: Weekly evidence across the original main comparison, added panel and earlier-period comparisons. Counts refer to related tasks rather than independent dealers. Stronger regularization narrows losses while leaving substantial failures.

Comparison set	Procedure	Loss change / median (%)	Higher-loss cells
Original Treasury-bill main comparison	Original regression, long history	+50.79	Original main comparison
Earlier Treasury-bill short-window version	Short-history short-window version	+144.30	Different history
Additional panel	Original full regression	+92.63	45 / 48
Additional panel	Four-feature regression	+9.91	37 / 48
Additional panel	Full regression, expanded penalties	+9.22	34 / 48
Additional panel	Ordinary empirical headroom quantile	+0.50	30 / 48
Earlier weekly groups	Validation-minimum expanded rule	+4.15	26 groups
Earlier weekly groups	One-standard-error rule	+5.37	26 groups

The original bill main comparison has 54 versus 28 strict breaches among 277 issues. Earlier short-history bill results use a different history; the one-standard-error worst excess is +181.39%, and all its prewritten success criteria remain unmet. Supplement Sections S11–S15 list all panel and selection results (Tables S9–S13).

Table 6: CFTC dealer-category futures: relative loss versus matched historical headroom (%). All 17 selected literal contract codes are retained.

Contract	History 52					History 260				
code	Intercept only	Four-feature	Full regression	Calibrated history	Calibrated regression	Intercept only	Four-feature	Full regression	Calibrated history	Calibrated regression
044601	5.26	128.00	41.38	15.01	18.80	0.51	7.82	43.88	13.87	42.41
043602	3.37	9.35	3.37	37.79	24.64	2.30	8.95	20.97	-11.22	-12.64
042601	4.64	9.48	35.65	26.69	41.21	-0.15	74.23	16.63	1.46	9.39
045601	-2.04	0.31	-2.04	21.83	6.14	1.85	1.85	1.85	2.40	2.58
020601	4.77	79.72	111.12	25.66	-5.34	0.92	43.38	29.70	-15.12	-12.32
099741	0.69	13.97	14.56	77.33	-13.28	0.22	0.22	0.22	27.72	27.73
1170E1	6.42	6.82	46.30	-0.22	12.77	0.92	-28.44	-16.22	-1.12	1.00
097741	-0.83	23.76	68.71	29.77	33.79	0.53	-6.54	14.69	-0.41	-5.43
096742	2.93	3.41	6.13	49.54	30.46	5.30	4.82	62.68	37.12	-0.49
095741	7.39	19.55	19.33	41.27	42.98	-0.70	2.13	10.35	19.30	55.39
090741	4.03	10.28	4.03	15.58	13.13	3.28	-5.35	-3.60	-2.02	-5.66
232741	0.11	123.33	37.53	63.84	36.35	1.94	-3.35	7.68	-0.65	-3.86
33874A	6.95	88.57	38.30	30.15	44.07	-2.38	-3.21	-4.78	6.98	9.52
240743	-3.04	-7.32	183.12	33.14	49.77	-2.28	-2.28	-2.28	32.91	26.37
092741	0.54	0.54	0.54	-1.74	-0.68	0.17	-6.98	26.22	5.58	13.94
098662	-0.44	-0.44	0.90	44.89	38.90	2.45	-39.44	2.45	15.12	20.45
112741	11.06	38.76	10.05	33.72	24.10	2.24	2.82	80.98	8.36	16.37

Entries are percentage loss differences from matched corrected historical headroom; positive values mean higher loss, and unavailable entries retain failed or incomplete comparisons. The pre-specified conditions require historical superiority in at least 60% of all 34 comparisons, full-regression medians within ten percent at both histories, and a strictly smaller four-feature median with longer history. Full-precision results determine the comparisons, with all scores, breaches, selections and failures retained in Supplement Section S30.

Table 7: Full-regression excess loss versus matched historical headroom under longer validation. Entries are source medians in percent; the final column counts selections of the intercept-only limit.

Reporting system	Horizon	History	Series	Validation 52	Validation 156	Limit selections
CFTC	6	52	17	+19.33	+4.03	4 to 10
CFTC	6	260	17	+10.35	+0.53	4 to 5
New York Fed	6	52	13	+6.68	-1.13	6 to 9
New York Fed	24	52	13	+40.47	+5.02	3 to 9

Lengths count weekly records; evaluation dates remain unchanged. New York Fed horizon six also moves the validation endpoint and refit phase; across-arm medians are not typical paired changes. Supplement Section S38 gives individual series, four-feature results and the complete figure.

Table 8: Full-regression relative excess loss (%) at alternative quantile levels. Positive values favor matched historical headroom; NA retains any failed required comparison.

Source / training history	90%	99%
CFTC, 52 weeks	+3.45	+15.05
CFTC, 260 weeks	+1.58	NA
FX, original period	-10.15	NA
FX, earlier period	-17.85	NA
Treasury bills	+0.67	+44.22

Contract rows give full-group medians; histories are 2,880 hours for FX and 260 weeks for Treasury. The rejected post-failure repair leaves original NA entries unchanged (Supplement Section S38).